

ISL Model Run Report (Fast Epoch 3)

Overview

This report summarizes the fast training run (3 epochs) and the comparison results for all four models: MobileNetV2, ResNet50, EfficientNetB0, and Custom CNN.

How Fast Epoch 3 Training Was Run

The runner script enables fast mode by default when no flags are passed. Fast mode sets the environment variables below for all training scripts:

- ISL_FAST_TRAIN=1
- ISL_FAST_EPOCHS=3

This run used fast mode with 3 epochs per model. You can reproduce it using:

```
python RUN_ALL_MODELS_AND_COMPARE.py --fast-epochs 3
```

Dataset Summary

- Data directory: Data
- Image size: 224 x 224
- Validation split: 0.2
- Training samples: 7216
- Validation samples: 1796

Results Table

Model	Epochs	Train Samples	Val Samples	Accuracy (%)	Loss	Parameters	Model Size (MB)	Inference Time (ms)
MobileNetV2	3	7216	1796	98.22	0.1243	2,621,009	24.81	18.42
ResNet50	3	7216	1796	14.25	3.4641	24,148,881	226.17	13.82
EfficientNetB0	3	7216	1796	7.46	2.7189	4,382,900	35.40	16.61
Custom CNN	3	7216	1796	52.34	3.3224	2,759,729	31.74	9.97

Result Files

- Comparison JSON: [Model/model_comparison_results.json](#)
- Run metadata: [Model/training_run_metadata.json](#)
- Comparison chart: [Model/model_comparison.png](#)

Comparison JSON (Tabular)

Model	Accuracy	Loss	Parameters	Model Size (MB)	Inference Time (ms)	Epochs	Train Samples	Val Samples
MobileNetV2	0.9822	0.1243	2,621,009	24.81	18.42	3	7216	1796
ResNet50	0.1425	3.4641	24,148,881	226.17	13.82	3	7216	1796
EfficientNetB0	0.0746	2.7189	4,382,900	35.40	16.61	3	7216	1796
Custom CNN	0.5234	3.3224	2,759,729	31.74	9.97	3	7216	1796

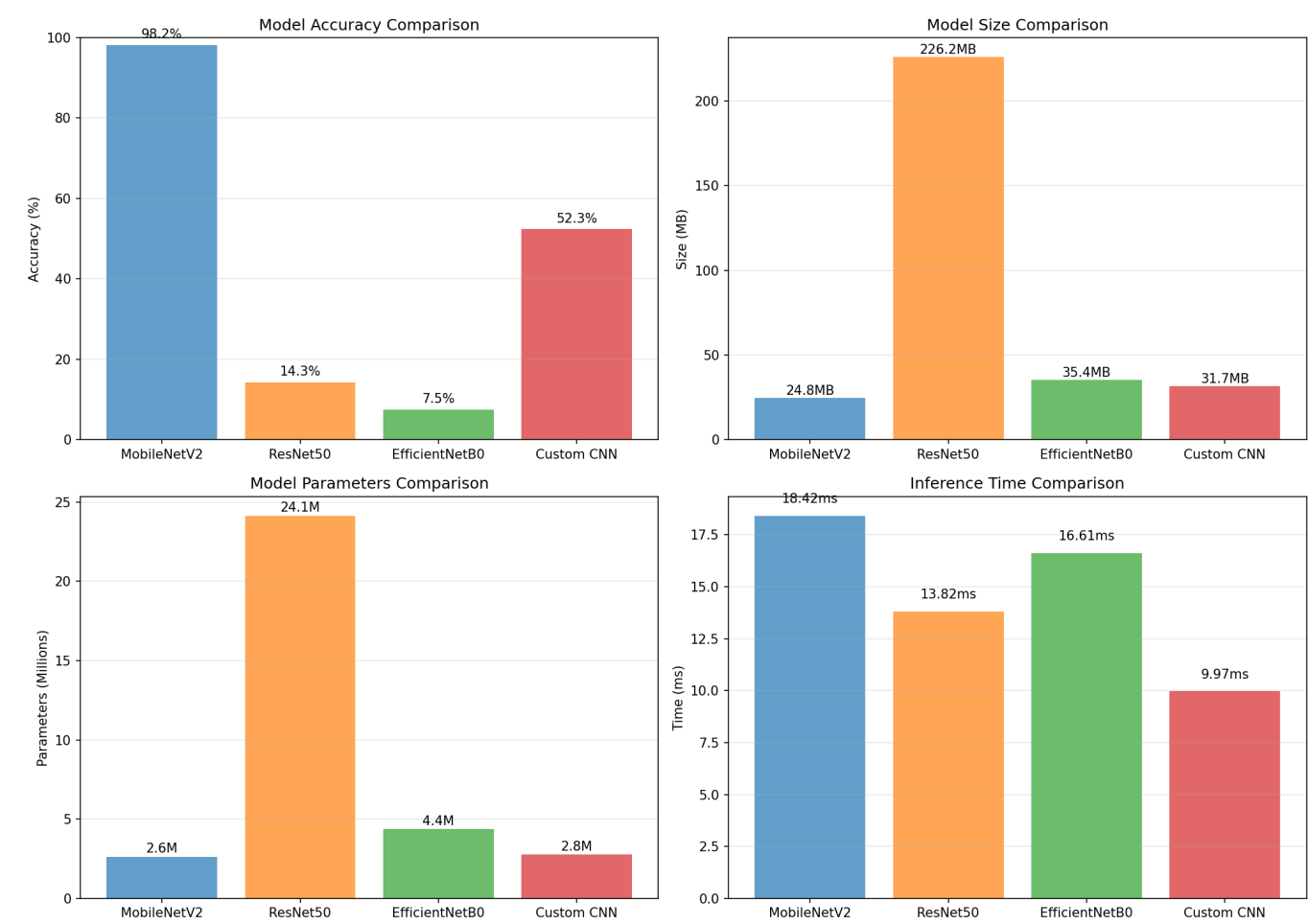
Training Run Metadata (Tabular)

Field	Value
Data directory	Data
Image size	224
Validation split	0.2

Model	Epochs	Train Samples	Val Samples
MobileNetV2	3	7216	1796
ResNet50	3	7216	1796
EfficientNetB0	3	7216	1796
Custom CNN	3	7216	1796

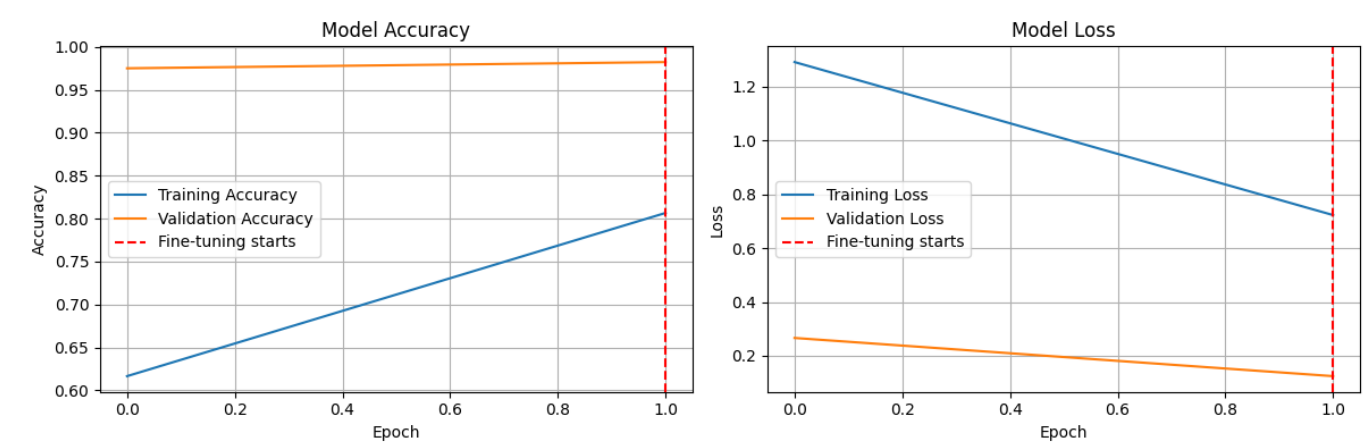
Charts

Comparison Summary

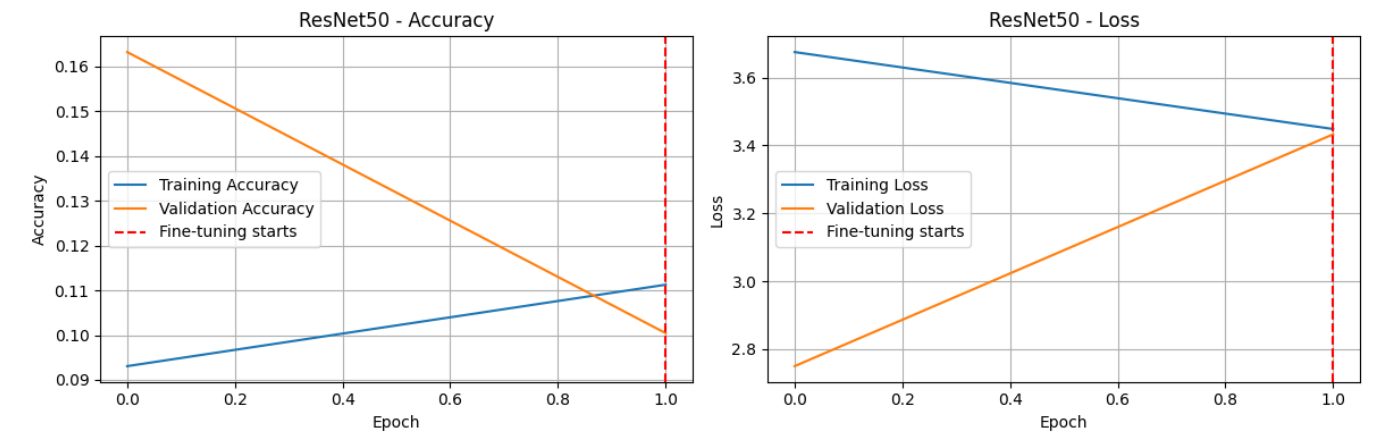


Training History (Per Model)

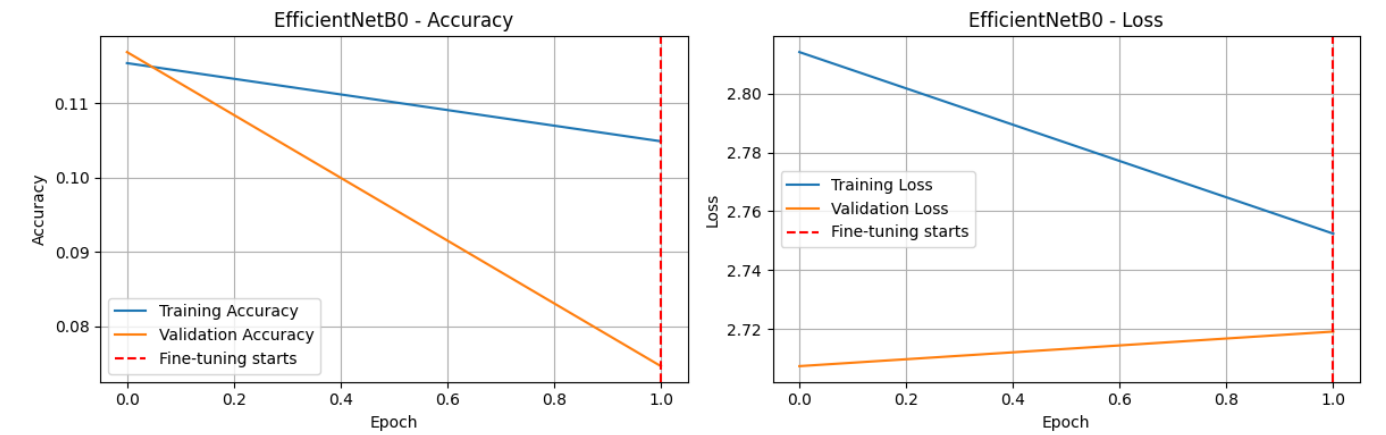
MobileNetV2



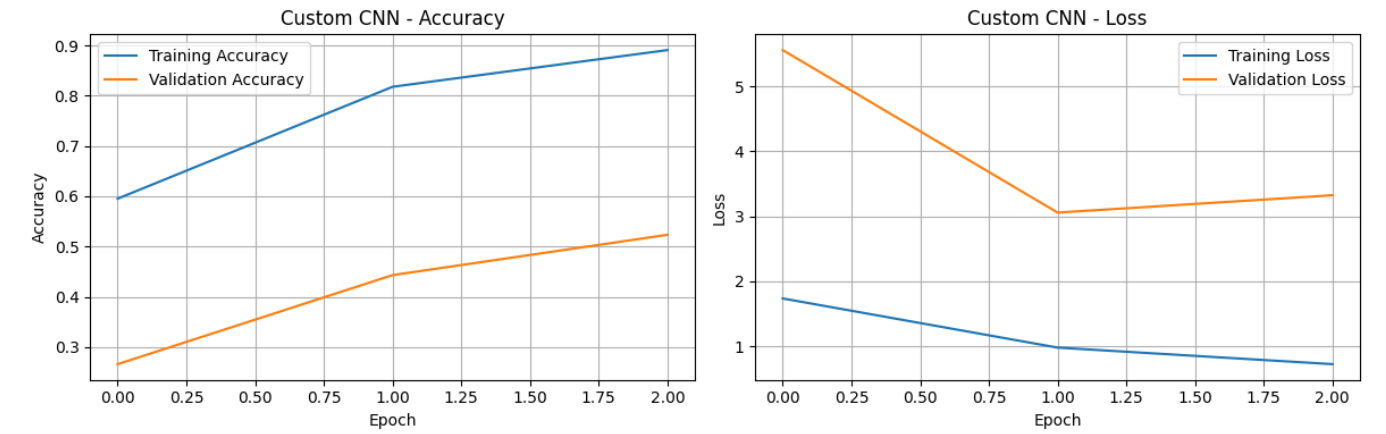
ResNet50



EfficientNetB0



Custom CNN



Notes

- Per-class accuracy details are stored in [Model/model_comparison_results.json](#).
- Fast mode is intended for quick comparisons and sanity checks; for higher accuracy, run full training with `--full` or raise epochs with `--epochs`.